

Guilherme Fiusa

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Research Interests Stochastic and quantum thermodynamics; thermodynamic inference from partial observations; full counting statistics; feedback control in open quantum systems

Personal Citizenship: Brazilian; currently residing in the USA

Education

University of Rochester, Rochester, USA
Ph.D. in Physics, 2023 – Present
Expected defense: May 2027
Advisor: Gabriel T. Landi

University of São Paulo, São Carlos, Brazil
M.Sc. in Physics, 2021 – 2022
Advisor: Diogo O. Soares-Pinto

University of São Paulo, São Carlos, Brazil
B.Sc. in Physics, 2018 – 2020
Advisor: Luiz Agostinho Ferreira

Positions

University of Rochester, Rochester, USA
Graduate Research Assistant, 2024 – Present
Teaching Assistant, 2023 – 2024

University of São Paulo, São Carlos, Brazil
Research Assistant, 2021 – 2022
Undergraduate Research Assistant, 2018 – 2020

Publications

Published

1. “Framework for fluctuating times and counting observables in stochastic excursions”. **G. Fiusa**, P. E. Harunari, A. S. Hegde, and G. T. Landi, *Phys. Rev. E* **114**, 024129 (2026).
2. “Counting observables in stochastic excursions”. **G. Fiusa**, P. E. Harunari, A. S. Hegde, and G. T. Landi, *Phys. Rev. E* **114**, L022102 (2026).
3. “Full counting statistics for jump-based feedback”. A. J. B. Rosal, **G. Fiusa**, P. Potts, and G. T. Landi, *Phys. Rev. A* **114**, 012216 (2026).
4. “Roadmap on Quantum Thermodynamics”. **G. Fiusa**, A. S. Hegde, and G. T. Landi (chapter contribution), *Quantum Sci. Technol.* **11**, 012501 (2026).
5. “Queued quantum collision models”. **G. Fiusa** and G. T. Landi, *Phys. Rev. A* **111**, 032220 (2025).
6. “Fidelity-based distance bounds for N -qubit approximate quantum error correction”. **G. Fiusa**, D. O. Soares-Pinto, and D. P. Pires, *Phys. Rev. A* **107**, 032422 (2023).

Preprints and manuscripts in preparation

1. “Stochastic trajectories and excursions in a double quantum dot system”. **G. Fiusa**, P. E. Harunari, A. J. B. Rosal, J. M. Nichol, and G. T. Landi, [arXiv:2605.20166](https://arxiv.org/abs/2605.20166) (2026).

2. “Thermodynamic inference from excursions into hidden states: theory and quantum-dot experiment”. **G. Fiusa**, J. van der Meer, T. V. Vu, V. Wadhia, G. T. Landi, and P. E. Harunari, in preparation (2026).
3. “Mutual linearity in zero-deficiency chemical reaction networks”. P. E. Harunari, **G. Fiusa**, and M. Polettini, in preparation (2026).

Theses

1. “Bounds for approximate quantum error correction and a few words about holography”. **G. Fiusa**, M.Sc. dissertation, University of São Paulo (2022). [\[link\]](#)
2. “Classical solutions of the SU(2) Yang–Mills theory”. **G. Fiusa**, B.Sc. thesis, University of São Paulo (2020). [\[link\]](#)

Seminars and Invited Talks

1. General Physics Seminars, University of São Paulo, São Paulo, Brazil, 2026
Invited talk: “Stochastic excursions: thermodynamics and inference”
2. II. Institute for Theoretical Physics, University of Stuttgart, Stuttgart, Germany, 2026
Invited talk: “Stochastic excursions: thermodynamics and inference”
3. Quantitative Life Sciences, ICTP, Trieste, Italy, 2026
Invited talk: “Stochastic excursions: thermodynamics and inference”
4. CENTURI, Aix-Marseille Université, Marseille, France, 2026
Invited talk: “Stochastic excursions: counting observables and inference”
5. São Carlos Institute of Physics, University of São Paulo, São Carlos, Brazil, 2026
Invited talk: “Thermodynamics of counting observables in stochastic excursions”
6. Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan, 2025
Invited talk: “Thermodynamics of counting observables in stochastic excursions”

Conferences

1. 2026 Quantum Thermodynamics
International Institute of Physics, Natal, Brazil, 2026
Contributed talk: “Stochastic excursions: thermodynamics and inference”
2. Workshop on Stochastic Thermodynamics (WOST)
Online, 2026
Contributed talk: “Stochastic excursions: thermodynamics and inference”
3. Maryland-Rochester Quantum Thermodynamics Symposium
University of Rochester, Rochester, USA, 2026
Poster: “Thermodynamics and counting observables in stochastic excursions”
4. Kyoto Workshop on Stochastic and Quantum Thermodynamics
Kyoto University & Yukawa Institute for Theoretical Physics, Kyoto, Japan, 2025
Poster: “Counting observables in stochastic excursions”
5. Rochester Conference on Coherence and Quantum Science
University of Rochester, Rochester, USA, 2025
Poster: “Counting observables in stochastic excursions”
6. Time in Quantum Theory 2025
University of Pavia & IFRN, Genoa, Italy, 2025
Contributed talk: “Time and thermodynamics of stochastic excursions”
7. III Workshop on Quantum Information and Thermodynamics
Brazilian Center for Research in Physics, Rio de Janeiro, Brazil, 2024
Poster: “Counting observables in stochastic excursions”

8. 2024 Quantum Thermodynamics
University of Maryland, College Park, USA, 2024
Poster: “Queued quantum collision models”
9. 2024 APS March Meeting
American Physical Society, Minneapolis, USA, 2024
Contributed talk: “Queued quantum collision models”
10. Quantum Thermodynamics: Quantum Many-Body Meets Thermodynamics
Harvard University, Cambridge, USA, 2024
Poster: “Queued quantum collision models”

Research Visits

University of Stuttgart, Stuttgart, Germany

June – July 2026
Host: Prof. Udo Seifert

ICTP, Trieste, Italy

June 2026
Host: Prof. Édgar Roldán

Aix-Marseille Université, Marseille, France

May 2026
Host: Dr. Pedro E. Harunari

Yukawa Institute for Theoretical Physics, Kyoto, Japan

November – December 2025
Host: Prof. Tan Van Vu

Awards

Fluctuation Theorem Poster Prize, 2026

Best poster award
Maryland-Rochester Quantum Thermodynamics Symposium

Tamor Physics and Astronomy Award, 2026

Travel grant awarded to outstanding graduate students
Department of Physics and Astronomy, University of Rochester

Tamor Physics and Astronomy Award, 2025

Travel grant awarded to outstanding graduate students
Department of Physics and Astronomy, University of Rochester

Tamor Physics and Astronomy Award, 2024

Travel grant awarded to outstanding graduate students
Department of Physics and Astronomy, University of Rochester

American Association of Physics Teachers Prize, 2024

Recognition of exceptional teaching assistants
American Association of Physics Teachers

Bernhard Gross Prize, 2020

Best performing student (highest GPA) in the B.Sc. in Physics
São Carlos Institute of Physics, University of São Paulo

Other Awards

Anna Louise Baker Award, 2026

Prize awarded to promising and aspiring music students
Eastman School of Music, University of Rochester

Anna Louise Baker Award, 2025

Prize awarded to promising and aspiring music students

Eastman School of Music, University of Rochester

Teaching and Mentoring

Teaching Assistant

University of Rochester, 2023 – 2024

- First-Year Mechanics, Honors (PHYS 141)
- Third-Year Advanced Electromagnetism (PHYS 237)

Undergraduate Research Supervision

Julia Munoz (Applied Mathematics major, University of Rochester), 2024 – 2025

- Project: Memory effects in quantum collision models

Graduate Mentorship

Department of Physics and Astronomy, University of Rochester, 2023 – 2025

- Mentored incoming and first-year graduate students

Extension Projects

Student Representative, Main Council

São Carlos Institute of Physics, University of São Paulo, 2019 – 2020

- Elected by the undergraduate student body

Head of the Chess Club

University of São Paulo, São Carlos, 2018 – 2022

- Organized competitions and raised funds

14th Brazilian National Week of Science and Technology

Institute of Mathematical Sciences, University of São Paulo, 2017

- Organized chess activities and tournaments, and taught the game

Languages

Portuguese (native)

English (fluent; TOEFL iBT 117/120)

Spanish (fluent)

Technical Skills

Programming: Julia, C++, Python, Fortran

Symbolic computation: Mathematica

Data analysis: hidden Markov inference on experimental trajectories

Typesetting: L^AT_EX

Piano: ESM level 4

References and contact

1. Gabriel T. Landi

Associate Professor, University of Rochester

Ph.D. thesis supervisor and research collaborator

gabriel.landi(at)rochester.edu

2. Diogo O. Soares-Pinto

Associate Professor, University of São Paulo

M.Sc. dissertation supervisor and research collaborator

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3. Pedro E. Harunari

Postdoctoral Fellow, Aix-Marseille Université

Research collaborator

pedroharunari(at)gmail.com